

Ex.No: 5	Implement the Digital Signature Algorithm (DSA).
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AIM:

ALGORITHM:

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CODE:

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import hashlib

def modular_inverse(number, modulus): return
    pow(number, -1, modulus)

def message_hash(message, q):
    hash_value = hashlib.sha256(message.encode("utf-8")).digest()
    hash_integer = int.from_bytes(hash_value, byteorder="big") return
    hash_integer % q

def generate_signature(message, p, q, g, private_key, k):
    hash_value = message_hash(message, q)
    r = pow(g, k, p) % q
    k_inverse = modular_inverse(k, q)
    s = (k_inverse * (hash_value + private_key * r)) % q return r,
    s, hash_value

def verify_signature(message, signature, p, q, g, public_key): r, s
    = signature
    if not (0 < r < q and 0 < s < q): return
        False, None, None, None
    hash_value = message_hash(message, q) w
    = modular_inverse(s, q)
    u1 = (hash_value * w) % q u2 =
    (r * w) % q
    v = ((pow(g, u1, p) * pow(public_key, u2, p)) % p) % q
    return v == r, v, u1, u2

def main():
    p = 23
    q = 11
```

```

g = 4
private_key = 3
public_key = pow(g, private_key, p) k =
7
message = input("Enter the message: ")
signature_r, signature_s, hash_value = generate_signature( message,
    p, q, g, private_key, k
)
print("\n----- DSA Parameters   ")
print("Prime Number (p) :", p)
print("Prime Divisor (q) :", q)
print("Generator (g) :", g)
print("\n----- Key Generation      ")
print("Private Key (x) :", private_key)
print("Public Key (y) :", public_key)
print("\n----- Signature Generation   ")
print("Message :", message) print("Message
Hash :", hash_value) print("Random
Number (k) :", k) print("Signature r :",
signature_r) print("Signature s :",
signature_s)
print("Digital Signature :", (signature_r, signature_s)) valid,
v, u1, u2 = verify_signature(
    message,
    (signature_r, signature_s), p,
    q, g, public_key
)

```

```

print("\n----- Signature Verification      ")

print("u1 :", u1)

print("u2 :", u2)

print("v :", v)

print("r :", signature_r) if
valid:

    print("\nSignature Status : VALID") else:

    print("\nSignature Status : INVALID") if
name_____=="_main_":

    main()

```

OUTPUT:

```

Enter the message: HELLO

----- DSA Parameters -----
Prime Number (p) : 23
Prime Divisor (q) : 11
Generator (g) : 4

----- Key Generation -----
Private Key (x) : 3
Public Key (y) : 18

----- Signature Generation -----
Message : HELLO
Message Hash : 7
Random Number (k) : 7
Signature r : 8
Signature s : 6
Digital Signature : (8, 6)

----- Signature Verification -----
u1 : 3
u2 : 5
v : 8
r : 8

Signature Status : VALID

```

RESULT: